Number of CPU Cores

Parallel processing across layers and functions
- PHY threads
- Scheduling threads
- MAC & RLC threads
- Packet IO threads
- CP Control threads
- UP Data threads
- UPF forwarding threads
- Logging
- Overhead

CPU Clock Speed

Faster cores can out perform number of cores

CPU Core Isolation

Some workloads should not share CPU Cores. Core isolation aids in more stable performance.
- PHY Processing
- Fronthaul packet handling
- DPDK poll mode threads
- PTP & Time sensitive processes

DU Workload

More Bandwidth = More PRBs and I/Q data processing

More Cells = More parallel signal processing

More UEs = More grant, HARQ processes, buffer handling, PRACH activity, and scheduling decisions

Throughput Workload

High throughput
- Increased packet movement and scheduling
- CU UP PDCP & GTP-C handling
- UPF forwarding and NAT load
- NIC queue
- Memory bandwidth and buffer queue

IoT UEs
- Context scaling

High data rate UEs
- Throughput scaling

Physical Split

Separating components between servers can isolate
- CPU Cycles
- Cache
- Memory Bandwidth
- NIC Bandwidth
- Interrupt handling
- PCIe lanes
- Timing margins

Storage

- Log heavy lab deployments
- Packet captures
- Metrics and databases
- Container image hosting
- Local analytics

O-RAN DEPLOYMENT ARCHITECTURE SCALING

How compute, workload, and functional splits impact system complexity

Lab — 1 Cell

Small Field — 2 cells

Large Field — 10's of cells

10's / 100's of cells

Physical Configuration

Simple → Complex

Physical Footprint | Cabling | Networking | Power Requirements | Compute Performance (Cores, RAM) | Design | Integration | Timing & Synchronization

Scalability

Static → Flexible

Network Expansion | Monitoring & Observability | Software Lifecycle Management | Fault Management | Configuration Management | Automation | Security

Throughtput & Capacity

Low → High

Signaling Overhead | User Plane Data | Expected Devices/Users | Device Types | Mobility | Data Type (UL or DL heavy) | Cell Density

Requirements

Minimal → Maximum

Capacity | Throughput | Bandwidth | Latency | Resilience | Mobility | Security | UE Power Saving | Spectrum Efficiency

Technology Integration

Minimal → Optimal

Network Slicing | MEC | Beamforming | MIMO | AI/ML | IAB | RedCap | NTN | V2X | mmWave | Non Standard Techologies and Features

Cost

Least Expensive → Most Expensive

Servers | Radios | Power | O&M | Cables & Adapters

Deployment Examples

**RU, PTP, etc. are assumed connections*

Full Consolidation

Server A: gNB DU, gNB CU, 5GC

Semi Consolidation

Server A: gNB DU, gNB CU

Server B: 5GC

Full Distribution

Server A: gNB DU

Server B: gNB CU

Server C: 5GC

Full Distribution - UPF Split

Server A: gNB DU

Server B: gNB CU

Server C: 5GC CP

Server D: UPF

Edge DU with Centralized CU and Core

Edge Server A: gNB DU

Edge Server A+1: gNB DU

Edge Server A+n: gNB DU

Server B: gNB CU (Regional)

Server C: 5GC (Regional)

Best For

Full Consolidation
- Lab
- Demonstration
- Training
- Development
- Portable
- Low UE count
- Low throughput
- Single cell
- Research
- Portability over performance

Semi Consolidation
- Private NW pilots
- Fieldable systems
- Compact but practical
- Moderate UE count
- Moderate throughput
- One or two cells
- Limited space & footprint
- Edge deployment needs

Full Distribution
- High performance
- Clean functional separation
- Scalable by role
- Production like testing
- High UE count
- Moderate to high throughput
- Multiple cells
- Producion realism
- Repeatable benchmarking

Full Distribution - UPF Split
- High throughput
- MEC
- UPF performance test
- Network slicing
- Data heavy networks
- Edge applications
- Packet processing experimentation

Edge DU with Centralized CU and Core
- Geographically separated radio sites
- Central network management
- Edge deployments
- Campus networks
- Private networks
- Central data center available
- Limited infrastructure at edge
- Expected NW growth

Pro's (+)

Full Consolidation
- Low cost
- Simple setup
- Minimal cabling
- Easy to clone

Semi Consolidation
- Low hardware count
- Simple F1 internal path
- Reduced inter-server traffic
- RAN & Core separation
- Cost vs. Scalability

Full Distribution
- Resource and component isolation
- Independent scaling
- Clean troubleshooting boundaries
- Fault isolation

Full Distribution - UPF Split
- Independent UP scaling
- UP data isolation

Edge DU with Centralized CU and Core
- Low latency fronthaul processing
- Central control and management
- Multi radio site scalability
- Reduced HW at edge sites
- Easily upgrade CU and Core

Con's (-)

Full Consolidation
- Resource contention
- Unrealistic benchmarking
- Weak fault isolation
- Limited Scaling

Semi Consolidation
- DU and CU resource competition
- Less isolated troubleshooting
- Lower growth headroom

Full Distribution
- High cost
- More NICs and cabling
- Complex network design
- More timing and interconnect
- Larger footprint

Full Distribution - UPF Split
- Additional costs
- Increased NW complexity and design
- Precise NIC and NUMA integration

Edge DU with Centralized CU and Core
- F1 latency and bandwidth requirements
- Reliable backhaul
- Most complex NW and design

NURADIO CONCEPTS